

The Rossmann Store Sales Forecasting Problem

A real-world data science case study

The real problem

Business question

Rossmann wants to forecast future daily sales for each store.

- Each row of the data represents one store on one day.
- The target variable is **Sales**: the daily turnover of that store.
- The aim is to predict future sales using historical information.

Can we predict how much money a store will take on a future day?

What data are we given?

Daily sales data

One row per store-day:

- Store
- Date
- DayOfWeek
- Open
- Promo
- StateHoliday
- SchoolHoliday
- Sales

Store description data

One row per store:

- StoreType
- Assortment
- CompetitionDistance
- Promo2
- PromoInterval

The two tables are joined using the Store column.

The forecasting setup

Training data

Historical rows where Sales is known.

Forecasting task

Future rows where Sales is hidden.

Store	Date	Open	Promo	SchoolHoliday	Sales
1	2015-06-30	1	1	0	5735
1	2015-07-01	1	1	0	?
1	2015-07-02	1	0	0	?

The model must learn from the past and forecast the future.

Our classroom version

Instead of using the official hidden Kaggle test set

We create our own test period from the labelled training data.

Period	Role	Sales column
Earlier dates	Model training	Visible
Final month	Test period	Hidden from the model
Final month	Evaluation	Revealed afterwards

We pretend the final month is the future.

Why feature engineering matters

Raw columns are not always enough

The model receives columns such as Date, Store, Promo and Holiday.

From these, we can create more useful features:

- Month
- Day of week
- Is the store open?
- Is there a promotion?
- Is this near a holiday?
- How far away is the nearest competitor?
- What were sales like recently?

Feature engineering means making hidden structure visible to the model.

Examples of engineered features

Raw information	Engineered feature	Why useful?
Date	Month	Captures seasonality
Date	DayOfWeek	Captures weekly patterns
Store history	SalesLag7	Same weekday last week
Store history	RollingMean28	Recent store trend
Promotion flag	Promo effect	Promotions affect demand
Store file	CompetitionDistance	Competitors may affect sales

Key idea

A good model needs good information. Feature engineering is how we create that information.

Baseline forecast

Before building a machine learning model, we need a simple benchmark.

Simple baseline

For each store, predict its average sales from the training period.

$$\widehat{\text{Sales}}_{\text{store}} = \text{average historical sales for that store}$$

A machine learning model is only useful if it beats a simple baseline.

The machine learning model

Model type

We use a regression model to predict a numerical value: daily Sales.

A strong model for this type of problem is:

XGBoost regression

It builds many decision trees, one after another.

- Early trees make rough predictions.
- Later trees correct previous mistakes.
- The final prediction combines many small corrections.

How do we judge the forecast?

The Rossmann competition used percentage-based error.

$$\text{RMSPE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{y_i} \right)^2}$$

- y_i is the true sales value.
- $\hat{y}_i$ is the predicted sales value.
- Lower RMSPE means better predictions.

This measures relative error, not just absolute error.

What the winning strategy teaches us

The winning Rossmann solution was not just about choosing a model.

Main ingredients

- Careful feature engineering
- Time-based validation
- XGBoost models
- Feature selection
- Ensembling several models

In real forecasting problems, the features often matter as much as the algorithm.

What we will do next

Workshop plan

In the remaining parts of the workshop, we will go through the full Rossmann forecasting process together using a Jupyter notebook.

We will use the notebook to:

- ➊ Load the Rossmann sales and store data.
- ➋ Join the tables using the Store column.
- ➌ Create our own test period by hiding the final month of Sales.
- ➍ Engineer useful features from dates, promotions, stores and recent sales.
- ➎ Build a simple baseline forecast.
- ➏ Train a machine learning model.
- ➐ Compare predictions with the true final-month sales.

The aim is to see the complete data science workflow, from raw data to forecast evaluation.